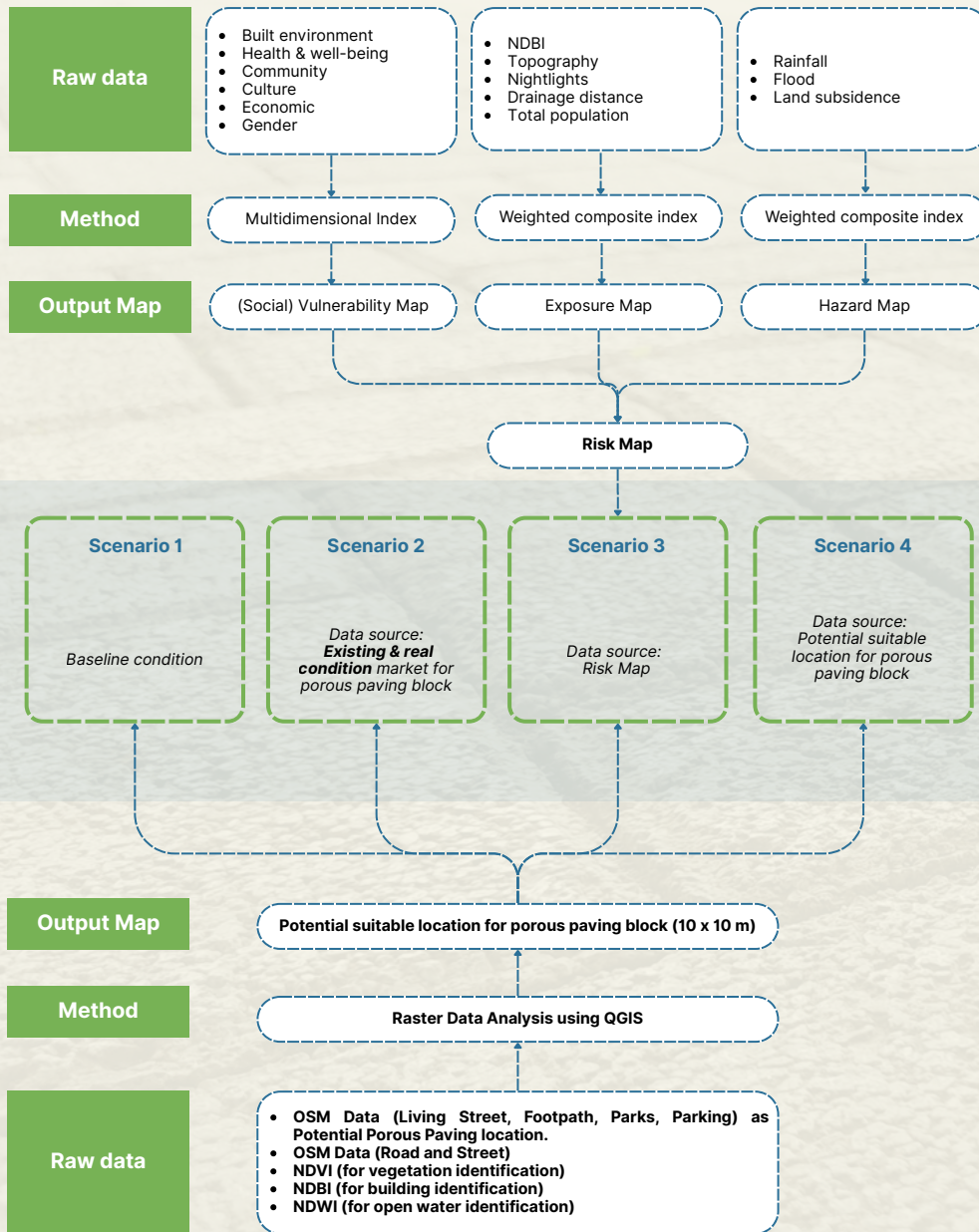


Decoded: For the Data Geeks



Sponge City for Everyone

*Spatial mapping to target
inclusive implementation*

Scan this QR code
for our **solution pack**:



**THE
YELLOW
SHEET**

- Haekal Akbar Kartasasmita
- Iqbal Fauzan Herlambang
- Sabrina Farah Salsabilla

Executive Summary

Jakarta's flood risk is rising—driven largely by land subsidence and land-use change. A practical response is the **Sponge City** approach: manage rain where it falls so it soaks in, is stored, or reused instead of rushing into drains and causing floods. One tangible measure is porous (permeable) paving, but today its adoption is still concentrated among **higher-income users**, limiting citywide impact.

In this project, **The Yellow Sheet** team formulated four scenarios which we tested using the runoff coefficient (C)—a 0–1 number showing the fraction of rainfall that becomes runoff (lower is better):

- 1 Current condition
- 2 Early adopters install porous paving block
- 3 Government targets high-risk areas
- 4 Citywide coverage on suitable surfaces

City-wide C is 0.733 today. **Early-adopter** action barely changes it (0.732). **Targeting high-risk areas** performs better (0.726). The strongest result comes from a full implementation, which lowers C to **0.696**—about a **5% reduction**—with decreases across all municipalities.

Jakarta can move from **concept to practice** by scaling porous paving blocks and other complementary measures. Our scenarios show that **depending to early adopter only changes little**, while **prioritizing high-risk areas delivers bigger results**, around 5% drop in city-wide run-off coefficient.

One pilot project in five sub-districts in high-risk area shows the **potential runoff reduction of 1 million m3 per year**. This can be the start for DKI Jakarta to widely implement by replicating pilot projects and tracking results with a simple SOP—with government, communities, and developers acting together.

Let's build a Sponge City for Everyone.

Jakarta is flooding & sinking, but how **worse** could it get?



Land subsidence due to groundwater extraction:

USD 421 million/year



Land use change due to less green space:

USD 270 million/year



Sea level rise due to climate change:

USD 212 million/year

Sponge City concept—an ongoing discussion

Sponge City is an urban development concept that manages rainwater like a **sponge**, instead of letting it quickly run-off into drains and cause floods.

absorb, store, filter, slowly release

The implementation includes but **not limited to**:



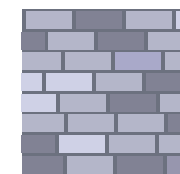
Urban planning & policy

Zoning regulation, incentives for low-impact development (LID)



Green infrastructure

Green roofs, rain gardens, bioswales, vegetated areas that absorb water



Permeable surfaces

Allowing rainwater to infiltrate into soil, e.g. porous paving block



This project focuses on this topic!



Porous paving block, with one of the widely implemented called PoreBlock, infiltrate water **100x times better** than conventional paving block.

Benefits

- Reduces **run-off load** on the drainage system
- Recharge **ground water**
- Reduces **surface temperature up to 30%** compare to conventional paving block

Technical features

- Concrete strength: **Class B**
- Compressive strength: **K-250**

Suitable location

- ✓ School
- ✓ Parking space
- ✓ City park
- ✓ Mosque / church
- ✓ Residential area

Porous paving blocks support the **Zero Run-off** principle by infiltrating rain on-site, so the development **does not increase runoff** to surrounding areas.

Meet them, **The Early Adopter** of porous paving block!



Bu Ari - Household
(38 years old)

She depends on **groundwater** for her family's daily needs. For her, porous paving block is a simple way to **recharge the groundwater** she relies on, while also being low-maintenance compared to other options.



Pak Fachmi - Entrepreneur
(42 years old)

He built a new eco-school and chose PoreBlock as part of his **water conservation strategy**. For him, it's not just infrastructure—it's also an **educational showcase** for children about how cities can manage water sustainably.

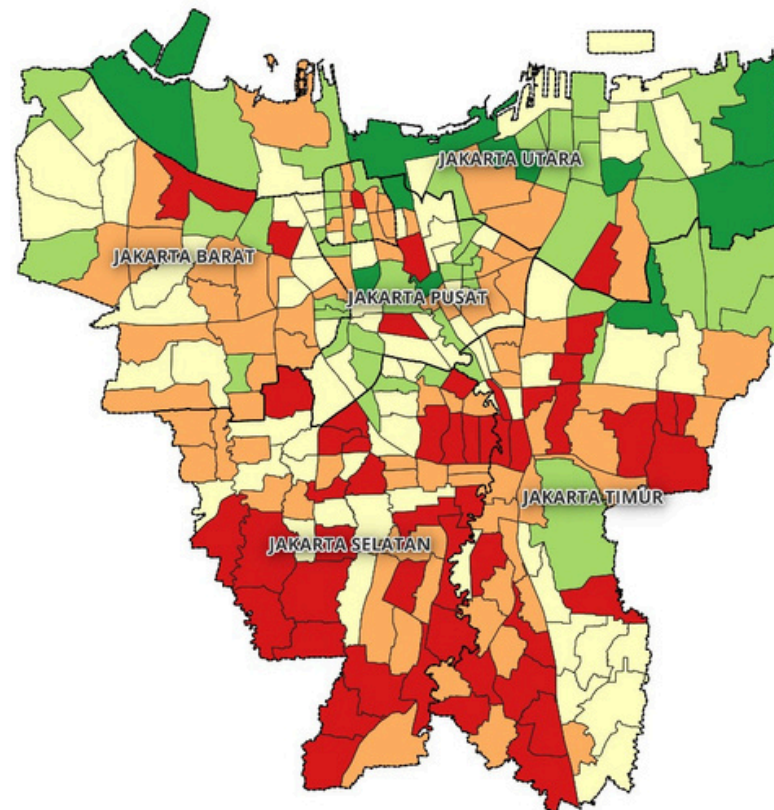


Pak Budi - The Architect
(55 years old)

With his professional background, he **understands concepts like Zero Run-off**. He installs porous paving block because he sees its long-term impact on reducing flooding and **improving urban water cycles**.

For the **sponge city concept** to work, solutions like porous paving blocks must be **implemented widely**; yet today, adoption is still confined to higher-income communities or people already aware of the problem, instead of being **inclusive** for everyone.

Based on our calculation of **Hazard, Risk, and (Social) Vulnerability** (please refer to the "Decode: For the Data Geeks" on the last page for detailed method), here's the **Risk Profile of DKI Jakarta**:



This map is to answer:

- ? Which locations to prioritize?
- ? How much is the area of the prioritization?
- ? Where are the **quick-win clusters**?

Imagine Jakarta as a **Sponge City--The What Ifs**

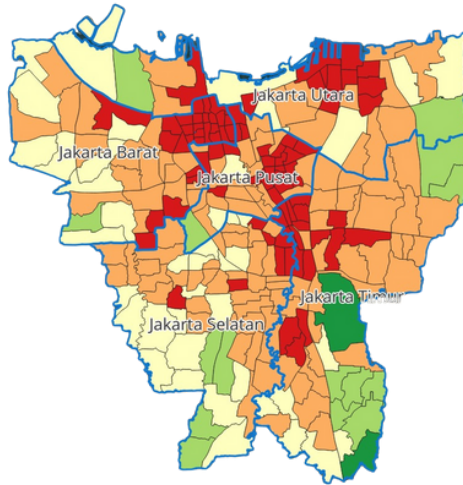
In this project, we use **run-off coefficient** as the main parameter.

Run-off coefficient (koefisien limpasan (C)) is a number from 0 to 1 showing **what fraction of rainfall leaves a surface**. The higher the number, the more rainfall leaves the site as runoff.

Example:

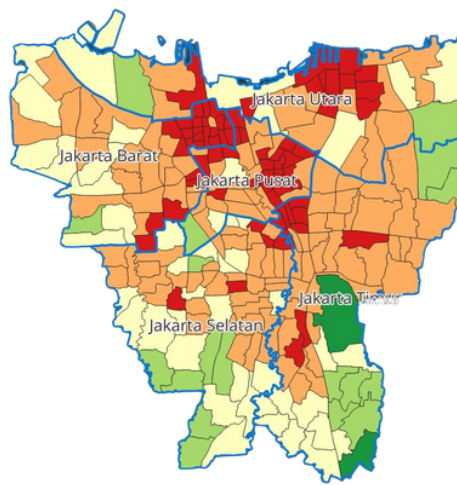
$C = 0.20$ means **20% of rainfall** becomes run-off.

Scenario 1: Current condition--how much runoff will occur when rain happens?

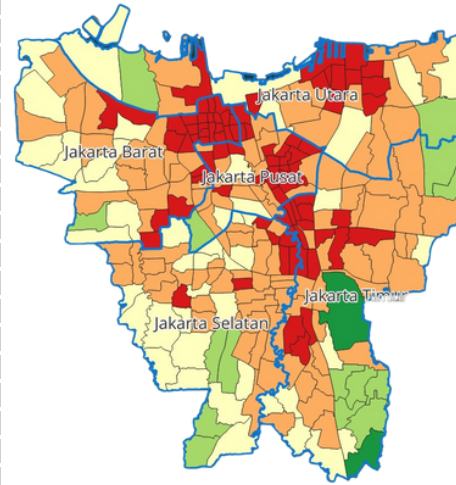


In current condition

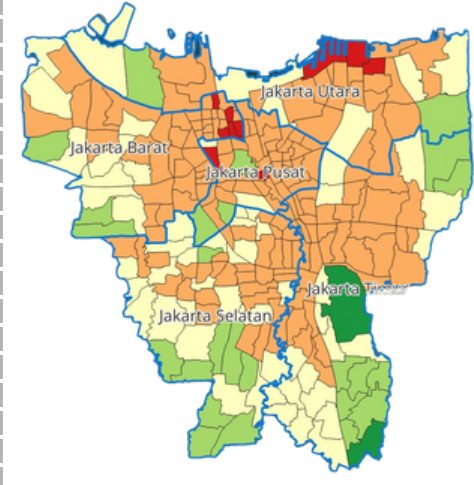
Scenario 2: What if people like **The Early Adopter** implement porous paving block?



Scenario 3: What if government intervenes so people in high-risk areas benefit?



Scenario 4: What if **all suitable sites** installed porous paving block?



North Jakarta

0.735

0.735

0.735

0.710

South Jakarta

0.694

0.689

0.677

0.659

West Jakarta

0.754

0.754

0.751

0.714

East Jakarta

0.697

0.696

0.682

0.652

Central Jakarta

0.786

0.784

0.784

0.747

DKI Jakarta

0.733

0.732

0.726

0.696

What does it mean to have runoff coefficient **decrease**?

Example:

0.04

decrease in Run-off Coefficient in **only one sub-district** (e.g. Tebet Timur) can result in:

100,000

m³ / year

run-off reduction

With scenario 1 to 4,

**Scenario 1:
Run-off coefficient**

0.733

**Scenario 4:
Run-off coefficient**

0.696

0.037

throughout DKI Jakarta,



Imagine how much runoff to the environment can be reduced every year

Co-benefits includes:

- ✓ Reduce time concentration (peak discharge)
- ✓ Increase Open Green Space (Ruang Terbuka Hijau/RTH)
- ✓ Recharging groundwater
- ✓ Implementation of nature-based solutions

Recommendations to **Strengthen** Implementation

Bappeda DKI Jakarta

Advance the **Zero Run-off principle** by scaling porous (permeable) paving in line with the DKI Jakarta **RPJMD 2025**, by these following tangible actions:

- **Mapping & prioritization**
Define priority zones & publish annual target maps
- **Technical standard**
C & time peak
- **Pilot & scale-up**
Execute pilot project at schools, sidewalks, public facilities, parking areas while setting annual targets (hectares & number of sites)
- **Financing & incentives**
Allocate regional budget, mobilize CSR/grants, partner with developers as private sector to build zero run-off system in newly built houses
- **Monitoring & reporting**
Apply a simple measurement SOP to report the runoff coefficient, permeable area installed, and m3 infiltrated per site as impact measurement.

RW or Communities & Developers

Expand adoption of porous paving block as neighborhood-scale disaster-resilience infrastructure **without shifting costs onto residents.**

- **Musrenbang (community planning)**
Propose porous pathway package (e.g. for parking, sidewalk, etc.) and integrate with drainage & other programs
- **Internal cross-subsidy**
For developers to apply internal cross-subsidy with the site plan, so that people can buy houses with already-implemented infrastructure for Sponge City.
- **Technical assistance**
Provide short O&M training for residents & offer simple guides

Best practice in Semarang to reduce urban flooding risk



1,500m²
porous paving block

- **53 %** Reducing **runoff** using porous paving block across a 1500 m² area
- **230** The community get **benefits** from the project
- **60 %** Increased **knowledge** about flood issues

Replication of best practice in Jakarta: Pilot Project



Refer to The Yellow Sheet Solution Pack to see more details regarding risk profile.

In Jakarta, these sub-districts are potential to be pilot projects due to:

- **Risk profile**
- Presence of **suitable locations**, showed by light green
- Greater potential for **public acceptance**

When it is implemented, the impact would be:

Run-off coefficient

0.834



0.787

Run-off avoided

1 million

m³ / year

There are potentials of porous block implementation of:

600,000 m²

If we could, we would...

Project limitation and what would have been done if time allows

- **Calculate the change in surface run-off & peak run-off**
Run-off calculation can be calculated with hydrologic-related software such as HEC-HMS.
- **Explore method for weight in risk calculation**
The weight for risk calculation can be pre-processed with methods such as AHP.
- **Calculate the effect on Open Green Space (RTH)**
As PoreBlock has been incorporated in ATR/BPN Ministerial Regulation 14/2022 for the Green Space Index.

Tabel 11. Nilai faktor elemen dalam perhitungan bonus elemen pembentuk RTH

NO.	ELEMEN RTH	FAKTOR ELEMEN RTH
1	Perkerasan tidak berpori seperti beton dan aspal (per 1 m ²)	0,00
2	Aspal berpori (per 1 m ²)	0,10
3	Paving block (per 1 m ²)	0,15
4	Beton berpori (per 1 m ²)	0,20
5	Paving berpori (per 1 m ²)	0,25
6	Taman dalam pot atau planter box garden (per 1 m ²)	0,40

- **Incorporate other possible infrastructures for Sponge City**
Using the same method, we can assess how infiltration wells, rainwater harvesting, and rain gardens reduce the runoff coefficient.
- **More accurate selection of Pilot Project recommendation**
The planned pilot project can be more accurate if we can visit the condition and measure directly the area.